Khwaja Yunus Ali University

School of Science and Engineering

B.Sc. in CSE & EEE

Lesson Plan

Course Title : Physics - I
Course Code : CSE 0533 1101 (CSE) & PHY 0533 1101 (EEE)

Term: Spring – 2024	Date: 17.01.2024	Time: 10.00 AM	Room: 327	Activity: Lecture 6

Topics to be discussed: Electric Current, Conductance, Specific Resistance, Current density, drift velocity

Target CLO & PLO : CLO2-3 & PLO 2 & 5

Learning Outcomes : At the end of the class the students will be able to:

- ❖ Define current and current density.
- ❖ Explain the term Drift velocity.
- ❖ Explain the concept of Resistance and Conductance.
- ❖ Describe the atomic view of Resistance
- ❖ Describe the concept of electromotive force

Methods of Teaching:

Mode/Activities	Time
Feedback of previous class	10min
Interactive discussion	10min
Lecture Presentation on Topics	30 min
Group activity	25 min
Feedback session	10 min

Assessment Strategy : Quizzes/Class tests, Class Participation

Sample Questions (MCQ):

1) Unit of current density is ………….
   a) C b) C/m c) A d) A/m e) A/m^2
2) Reciprocal of the resistivity is called …………….
   a) resistance b) conductance c) capacitance d) conductivity e) permittivity

Sample Questions (SAQ):

1) What is current and current density?
2) What is the drift velocity? Drive the equation for it.
3) State and Explain Laws of resistance.
4) Define Resistance, resistivity, conductance, conductivity.
5) State and Explain Ohm's Law.
6) Describe the atomic view of Resistance.
7) Define electromotive force (e.m.f) and internal resistance. Deduce the expression for them.
7) What is the drift velocity of electrons in a copper conductor having a cross-sectional area of $5 \times 10^{-6}\ m^2$, if the current is 10A? Assume that there are 8×10^{28} electrons/m^3($q_e = -1.6 \times 10^{-19} c$).
8) A silver wire 1 mm in diameter carries a charge of 90 coulomb in 1 hr. Silver contains 508×10^{28} free electrons/m^3. Calculate (i) the current in the wire (ii) current density and drift velocity of electrons in the wire.

Reference Books:
1. Giasuddin ahmed – Physics for Engineers (Part-2)
2. Huq. Rafiqullah- Electricity and Magnetism

Pre-reading book for next Lecture:
- Dr. Amir Hussain Khan – Physics 2nd Paper (HSC Text Book), Pages: 101-105
- Giasuddin ahmed – Physics for Engineers (Part-2); Pages: 151-166